

A Counterexample to the Hardy-Space Norm Conjecture

Conjecture 4.3 of arXiv:2511.08000

Automated mathematical discovery run `fa_banach_001`
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Status: candidate counterexample, likely valid. The argument is exact: it uses finite rational coefficient sums and no numerical integration.

1 Source and target

The source is C. Bénéteau, R. Cheng, C. Felder, D. Khavinson, M. Manolaki, and K. Maronikolakis, *Metric projections, zeros of optimal polynomial approximants, and some extremal problems in Hardy spaces*, arXiv:2511.08000v2, Conjecture 4.3 on page 12 [1].

In light of Proposition [4.1](#) we conjecture the following:

Conjecture 4.3. Let $1 < p < \infty$ and $f \in H^p$ with $f(0) \neq 0$. If J is any non-constant inner function, then there exists a constant $c = c_f \in \mathbb{T}$ such that

$$\|1 - Jf\|_p > \|1 - cf\|_p.$$

By continuity, we can at least say that the desired inequality still holds for values of p which are close to two. In particular, using [\[20\]](#) Lemma 3.1.1], we obtain the following:

Figure 1: Conjecture 4.3 in the source paper, page 12.

With normalized Lebesgue measure m on $\mathbb{T}$, the conjecture reads:

Let $1 < p < \infty$ and $f \in H^p$ with $f(0) \neq 0$. If J is any nonconstant inner function, then there exists $c = c_f \in \mathbb{T}$ such that

$$\|1 - Jf\|_p > \|1 - cf\|_p.$$

2 Idea of the proof

The exponent $p = 4$ makes the proposed inequality completely algebraic: for every analytic $h \in H^4$, Parseval gives $\|h\|_4^4 = \|h^2\|_2^2$. We choose a short real polynomial f . The quantity $\|1 - cf\|_4^4$, for $|c| = 1$, then becomes a quadratic polynomial in $x = \operatorname{Re} c$; its global minimum is read off on $[-1, 1]$. For the left side we choose a single Blaschke factor J . Multiplication by J is isometric in H^2 , so only one finite coefficient pairing remains. The two exact rational values occur in the strict reverse order.

3 Counterexample

Theorem 1. *Set*

$$p = 4, \quad f(z) = \frac{1}{2} - 3z + 2z^3 - z^5, \quad J(z) = \frac{z - \frac{1}{4}}{1 - \frac{1}{4}z}.$$

Then $f(0) = \frac{1}{2}$, and J is a nonconstant inner function with $J(0) = -\frac{1}{4}$. Moreover,

$$\|1 - Jf\|_4^4 = \frac{1344927}{4096} < \frac{5697}{16} = \min_{|c|=1} \|1 - cf\|_4^4.$$

Consequently, every $c \in \mathbb{T}$ satisfies

$$\|1 - Jf\|_4 < \|1 - cf\|_4,$$

so Conjecture 4.3 is false.

Proof. All inner products and norms below are taken on $\mathbb{T}$ with normalized Lebesgue measure. If $h(z) = \sum_{n \geq 0} h_n z^n \in H^4$, then

$$\|h\|_4^4 = \int_{\mathbb{T}} |h|^4 dm = \|h^2\|_2^2 = \sum_{n \geq 0} |\widehat{h^2}(n)|^2. \quad (1)$$

The constant-phase side. For $|c| = 1$, direct expansion gives the nonzero coefficients of $(1 - cf)^2$:

n	0	1	2	3	4	5	6	8	10
$[z^n](1 - cf)^2$	$\frac{(c-2)^2}{4}$	$-3c(c-2)$	$9c^2$	$2c(c-2)$	$-12c^2$	$-c(c-2)$	$10c^2$	$-4c^2$	c^2

Applying (1) and writing $c = e^{i\theta}$ and $x = \cos \theta$, we obtain

$$\begin{aligned} \|1 - cf\|_4^4 &= \frac{6625}{16} + \frac{1}{2} \cos(2\theta) - \frac{117}{2} \cos \theta \\ &= \frac{6617}{16} + x^2 - \frac{117}{2} x. \end{aligned} \quad (2)$$

The derivative $2x - \frac{117}{2}$ is negative throughout $[-1, 1]$. Thus (2) is minimized at $x = 1$, and

$$\min_{|c|=1} \|1 - cf\|_4^4 = \frac{6617}{16} + 1 - \frac{117}{2} = \frac{5697}{16}. \quad (3)$$

The inner-function side. Put $u = Jf$. Expanding $(1 - u)^2 = 1 - 2u + u^2$ in H^2 gives

$$\|1 - Jf\|_4^4 = 1 + 4\|f\|_2^2 + \|f^2\|_2^2 - 4\operatorname{Re}((Jf)(0)) + 2\operatorname{Re}((Jf)(0)^2) - 4\operatorname{Re}\langle f, Jf^2 \rangle. \quad (4)$$

Here we used that multiplication by the inner function J is an isometry on H^2 , and hence

$$\langle Jf, J^2 f^2 \rangle = \langle f, Jf^2 \rangle.$$

Finite coefficient sums yield

$$\|f\|_2^2 = \frac{57}{4}, \quad \|f^2\|_2^2 = \frac{5697}{16}.$$

For completeness, the Taylor polynomial through degree five of Jf^2 is

$$Jf^2 = -\frac{1}{16} + \frac{63}{64}z - \frac{1281}{256}z^2 + \frac{7423}{1024}z^3 + \frac{27903}{4096}z^4 - \frac{164609}{16384}z^5 + O(z^6).$$

Since f has degree five, this is enough to compute the mixed term:

$$\langle f, Jf^2 \rangle = \frac{353249}{16384}.$$

Finally, $(Jf)(0) = -\frac{1}{8}$. Substitution into (4) gives

$$\begin{aligned} \|1 - Jf\|_4^4 &= \frac{6625}{16} + \frac{1}{2} + \frac{1}{32} - 4 \left(\frac{353249}{16384} \right) \\ &= \frac{1344927}{4096}. \end{aligned}$$

Comparison with (3) is strict:

$$\frac{5697}{16} - \frac{1344927}{4096} = \frac{113505}{4096} > 0.$$

Taking fourth roots preserves the inequality and proves the claim. $\square$

4 Verification

The script `code/verify_counterexample.py` recomputes every displayed quantity using Python's exact `Fraction` arithmetic. It uses the Taylor formula

$$[z^0]J = -a, \quad [z^n]J = (1 - a^2)a^{n-1} \quad (n \geq 1), \quad a = \frac{1}{4},$$

and polynomial convolution; it performs no quadrature or floating-point comparison. The command

```
python code/verify_counterexample.py
```

returns the two exact fourth powers, their positive gap, and a passing status.

5 Scope and novelty check

This example disproves Conjecture 4.3 exactly, and it also satisfies the stronger side condition $J(0) \neq 0$ appearing in the preceding Proposition 4.2 of the source. It does *not* by itself construct an optimal polynomial approximant with a zero in $\mathbb{D}$: Proposition 4.2 is a one-way sufficient mechanism, so the broader zero-free OPA question is not settled here.

Before promotion, the run's lightweight indexes were searched for arXiv:2511.08000, the exact title, the Hardy-space norm inequality, inner functions, and OPA zeros. Bounded web searches on 23 July 2026 checked the current arXiv v2 record, the Canadian Journal of Mathematics publication record, the exact conjecture phrase and label, and close Hardy-space/optimal-polynomial-approximant terms. These searches found the source and background OPA papers but no later proof, counterexample, correction, or explicit resolution of Conjecture 4.3. Novelty is therefore plausible but remains subject to expert literature review.

6 Human-review recommendation

The counterexample is recommended for mathematical review as *likely valid*. The most useful manual check is the mixed coefficient pairing $\langle f, Jf^2 \rangle = 353249/16384$; the exact verifier independently reconstructs it from polynomial and Blaschke coefficients.

Exact verifier report

Running the bundled verifier in the packet directory returns:

```
||1-Jf||_4^4 = 1344927/4096
inf_|c|=1 ||1-cf||_4^4 = 5697/16
strict reversed gap = 113505/4096
PASS: Conjecture 4.3 is violated for every unimodular c.
```

Suggested review checks

1. Confirm that $J(z) = (z - \frac{1}{4})/(1 - \frac{1}{4}z)$ is inner and $J(0) \neq 0$.
2. Re-sum the nine coefficients of $(1 - cf)^2$ and the quadratic in $x = \operatorname{Re} c$.
3. Recompute the six Taylor coefficients of Jf^2 used in the finite inner product.
4. Note that the conclusion is the strict reverse for every $c \in \mathbb{T}$, so no issue of phase selection remains.

References

- [1] C. Bénéteau, R. Cheng, C. Felder, D. Khavinson, M. Manolaki, and K. Maronikolakis, *Metric projections, zeros of optimal polynomial approximants, and some extremal problems in Hardy spaces*, arXiv:2511.08000v2, 2026.